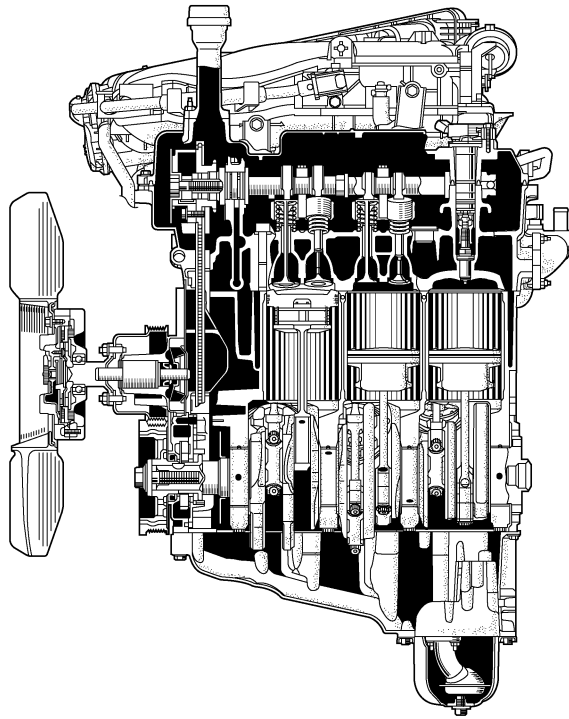


ENGINE

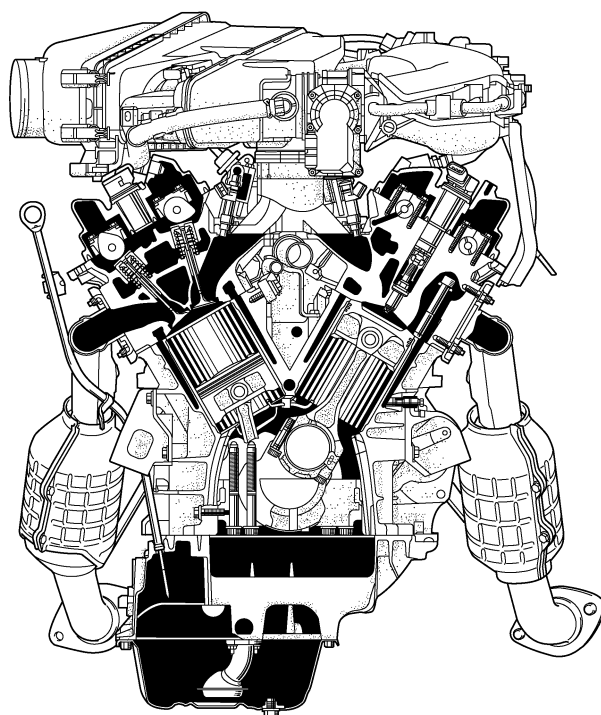
1GR-FE ENGINE

■ DESCRIPTION

The 1GR-FE engine is a V6 4.0-liter, 24-valve DOHC. This engine uses the VVT-i (Variable Valve Timing-intelligent) system, DIS (Direct Ignition System), ACIS (Acoustic Control Induction System), and ETCS-i (Electronic Throttle Control System-intelligent). These control functions achieve improved engine performance, fuel economy, and clean emissions.



238EG01



081EG06Y

► Engine Specifications ◀

No. of Cyls. & Arrangement			6-Cylinder, V Type
Valve Mechanism			24-Valve DOHC, Chain Drive (with VVT-i)
Combustion Chamber			Pentroof Type
Manifolds			Cross-flow
Fuel System			EFI
Ignition System			DIS
Displacement		cm ³ (cu. in.)	3956 (241.4)
Bore × Stroke		mm (in.)	94.0 × 95.0 (3.70 × 3.74)
Compression Ratio			10.0 : 1
Max. Output		[EEC-NET]	179 kW @ 5200 rpm
Max. Torque		[EEC-NET]	376 N·m @ 3800 rpm
Valve Timing	Intake	Open	42° BTDC to 8° ATDC
		Close	10° ABDC to 60° ABDC
	Exhaust	Open	54° BBDC
		Close	2° ATDC
Firing Order			1-2-3-4-5-6
Oil Grade			API grade SL “Energy-Conserving”, SM “Energy-Conserving” or ILSAC
Research Octane Number (RON)			95 or higher* ²
Emission Regulation	Models for G.C.C. Countries		EURO II
	Models except G.C.C. Countries		EURO III
Engine Service Mass* ¹ (Reference)		A/T	166 (366)
		M/T	181 (399)

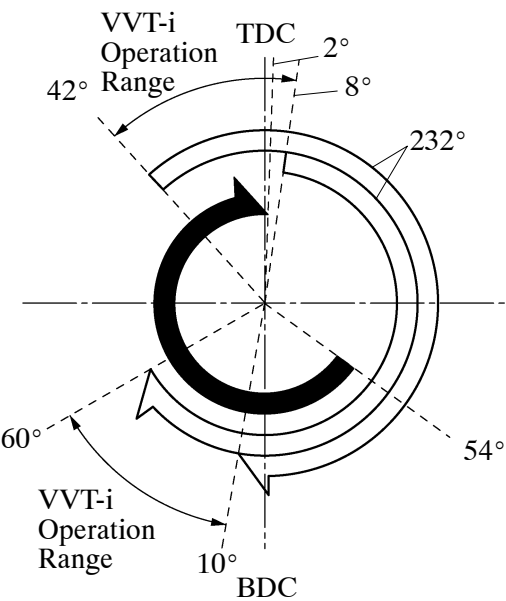
*1: The figure shown is the weight of the part without coolant and oil.

*2: Premium unleaded gasoline with a Research Octane Number 95 or higher required for optimum engine performance. If 95 octane cannot be obtained, you may use unleaded gasoline with a Research Octane Number as low as 91. Use of unleaded fuel with a Research Octane Number lower than 95 may result in engine knocking and significantly reduced performance. Persistent knocking can lead to engine damage and should be corrected by refueling with higher octane unleaded gasoline.

► Valve Timing ◀

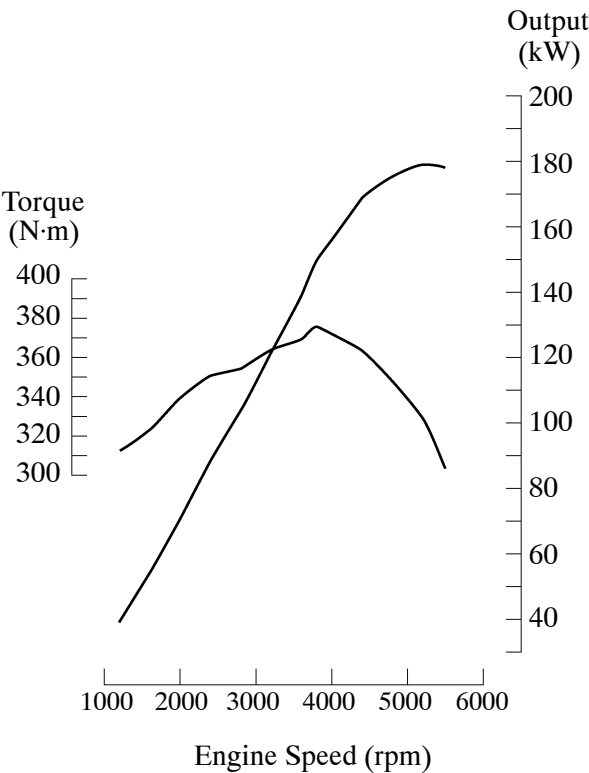
◁ : Intake Valve Opening Angle

▴ : Exhaust Valve Opening Angle



238EG04

► Performance Curve ◀



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